PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Correlation between masked hypertension and endothelial dysfunction measured by flow-mediated dilation：a protocol of systematic review and meta-analysis
AUTHORS	Zhang, Xiaohua; Li, Yong; Gao, Ling; Yu, Qian; Zhou, Congliang; zou, wenshu

VERSION 1 – REVIEW

REVIEWER	Higashi, Yukihito Hiroshima University, Department of Cardiovascular Regeneration and Medicine
REVIEW RETURNED	11-Feb-2021

GENERAL COMMENTS	This protocol paper is somewhat superficial and lacks a novelty what the authors want to emphasize. In addition, I am afraid the small number of published papers that showed the relationship between masked hypertension and flow-mediated dilation to perform a meta-analysis.

REVIEWER	Anpalahan, Mahesan University of Melbourne Faculty of Medicine Dentistry and Health Sciences, Medicien
REVIEW RETURNED	16-Mar-2021

GENERAL COMMENTS	The authors describe a protocol for a systematic review and meta-analysis that aims to determine the association between brachial flow mediated dilation (FMD), a measure of endothelial dysfunction, and masked hypertension. Although the overall search strategy and statistical analyses appear appropriate, the authors may wish to address the following. Major concerns: 1) It is not clear whether a distinction will be made between masked hypertension and masked uncontrolled hypertension. I believe this is important as this will have implications for the study methodology. 2) The protocol is extremely difficult to understand because of grammatical errors and poor sentence structuring. I believe the manuscript needs significant language editing in order to improve readability. 3) Although the exclusion criteria are clear, the inclusion or eligibility criteria are difficult to understand and need to be defined clearly. Minor concerns:

	Background: Perhaps the background should focus on the possible association between FMD and cardiovascular outcomes or masked hypertension / masked uncontrolled hypertension. The details on the clinical significance of hypertension and diagnosis of masked hypertension (the first two paragraphs of background) are probably not relevant to the research question.

REVIEWER	Siddiqui, Mohammed The University of Alabama at Birmingham, Medicine
REVIEW RETURNED	29-Mar-2021

GENERAL COMMENTS	The authors describe proposed meta-analysis of the observational studies to study the correlation between masked hypertension and endothelial dysfunction measured using flow-mediated dilation. I have few comments in methodology described in both abstract and methods section: 1. The authors should add search keyword masked uncontrolled hypertension, which is the terminology for masked hypertension on antihypertensive medications. 2. The eligibility criteria fails to mention people who are controlled in clinic as masked hypertension is defined as BP controlled in clinic and uncontrolled out-of-clinic by 24-hr ambulatory or home BP monitoring.

VERSION 1 – AUTHOR RESPONSE

Reply to Reviewer 1,
Dear Prof. Yukihito Higashi,
Thank you for your valuable comments. Although masked hypertension has overall cardiovascular risk equivalence to Stage-1 hypertension, the former is easier to be neglected. Per the convinence of FMD diagnosis, some researches were conducted to see if FMD is available to identify the risk. However, there is no related meta-analysis to provide higher quality evidence. Two electronic databases (PubMed, CNKI) were roughly searched (from January 2010 till now) for original articles, finding that there are at least nine articles including the correlation between masked hypertension and endothelial dysfunction measured using flow-mediated dilation.
Five selected articles in Pubmed are from Kabutoya T et al, published in J Clin Hypertens (Greenwich) in 2013(DOI:10.1111/jch.12155), Veerabhadrappa P et al, published in Am J Hypertens in 2011(DOI: 10.1038/ajh.2011.103), Sincer I et al, published in Clin Exp Hypertens in 2014(DOI: 10.3109/10641963.2013.827694), Caliskan M et al, published in Clin Exp Hypertens in 2012 (DOI: 10.3109/10641963.2012.712176), and Fekete AA et al, published in Am J Clin Nutr in 2016(DOI: 10.3945/ajcn.116.137919)
Other four selected articles in CNKI are from Yiyin Zheng et al, published in Heilongjiang Medical Journal in 2019, Tiantian Ji et al, published in Chinese Journal of Geriatric Cardio-Cerebrovascular Disease in 2018, Shenfa Fu et al, published in Chinese Journal of Hypertension in 2010, and Xin

Zhang et al, published in Journal of Xinxiang Medical College in 2013. We believe there are more than 10 articles after we implement a more careful search plan, which is sufficient to conduct a meta-analysis.

Reply to Reviewer 2,
Dear Mahesan Anpalahan,
Thank you for your valuable comment. We specified the inclusion criteria to include patients with masked hypertension as well as patients with masked uncontrolled hypertension. Although, from a rough search, more articles may refer to masked hypertension identify by its explicit diagnostic criteria, we will specifically distinguish upon the outcomes between MH and MUCH. We also modified the vocabulary and grammar of the article to avoid the mistakes caused by the language structure. The inclusion criteria are adults (above 18 years old, without upper age limit); Recessive hypertension was well defined (normal office blood pressure (BP<135/85 mmHg)(controlled or not), daytime or out-of-office home blood pressure > 135/85 mmHg, 24-hour blood pressure greater than 130/80 mmHg).We have modified the backgroud to adhere it to the theme.

Dear Mohammed Siddiqui,
Thank you for your valuable comments. We add search keyword masked uncontrolled hypertension, which is the terminology for masked hypertension on antihypertensive medications. We have modified the inclusion criteria, which can be seen from lines 147 to 150 on page 4 of the paper. The inclusion criteria are adults (above 18 years old, without upper age limit); Recessive hypertension was well defined (normal office blood pressure (BP<135/85 mmHg)(controlled or not), daytime or out-of-office home blood pressure > 135/85 mmHg, 24-hour blood pressure greater than 130/80 mmHg). When we collect the original articles, a specific search plan will be implemented to distinguish upon the outcomes between MH and MUCH.

VERSION 2 – REVIEW

REVIEWER	Anpalahan, Mahesan University of Melbourne Faculty of Medicine Dentistry and Health Sciences, Medicien
REVIEW RETURNED	09-Jul-2021

GENERAL COMMENTS	The conventional office BP threshold that defines normal office BP is <140/90, although in patients with specific comorbidities such as CKD or diabetes it may be defined at a lower threshold, for example 130/80. It is unclear why authors have chosen an office BP threshold <135/85. What do authors mean by" recessive hypertension"?

VERSION 2 – AUTHOR RESPONSE

Reviewer: 2
Dr. Mahesan Anpalahan, University of Melbourne Faculty of Medicine Dentistry and Health Sciences
Comments to the Author:
The conventional office BP threshold that defines normal office BP is <140/90, although in patients with specific comorbidities such as CKD or diabetes it may be defined at a lower threshold, for example 130/80. It is unclear why authors have chosen an office BP threshold <135/85.
Re: Thanks for your careful checking. I refer to diagnostic criteria of masked hypertension in some related articles. ffice BP <140/90 mmHg and home BP ≥135 mmHg and/or 85 mmHg is indeed more reasonable. I have revised the manuscript accordingly.
What do authors mean by" recessive hypertension"?

Re: I didn't find the expression in the manuscript. In my impression, I once used "recessive hypertension" due to the habit of the Chinese language, which is the same meaning of "masked hypertension". I will revise it if I see this expression.